

Some pages covering:

Mathematical statistics:

Accuracy measures

Error propagation law

Applications of the law

**For use in the course
TBA4231 APPLIED GEOMATICS**

Section of Geomatics
Department of transport and building science
NTNU

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Contents of this note:

This note is covering mathematical statistics as this topic partly was lectured in the course Applied geomatics. This note is an extension of the same topic in chapter 9 in the textbook "Surveying for Engineers". The introduction of co-variances and the use of matrices in the error propagation law are the main differences compared to the textbook.

See also the example exercises on It's learning, and the (voluntary) assignment no 8 covering the themes in this note.

This note is originally written in Norwegian as a compendium covering mathematical statistic and estimation. The numbers of the chapters are the same as in the original compendium.

This is the "beta" version, and the quality control of the text and the spelling could probably have been better.

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Terje Skogseth

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PART A: MATHEMATICAL STATISTICS

2. Accuracy measures. Standard deviation

See chapter 9 in the text book "Surveying for Engineers" for an introduction to this topic, especially about the definition of standard deviation.

3. Measurements of different accuracy. Weight.

3.1 The concept weight

We use the observables to calculate derived parameters like coordinates and heights. These observables normally have different accuracy.

In land surveying we normally are measuring more than "necessary": The number of observations exceed the number of observations that are necessary to determine the unknown of the system. In this case we carry out an estimation to determine the most probably values of the unknowns. In the estimation process the observations adjust to each other, they are as a result of the estimation corrected (adjusted observations).

We have to carry out the adjustments in such a way that the observables with high accuracy also are given a relatively strong influence on the results. We therefore must give the observations different weights according to the accuracy of the particular observations.

The accuracy of an observable is characterized by its standard deviation. The standard deviation has earlier been used as an accuracy measure of the observables and variables estimated from the observables. The weight, p , for a particular observable must therefore have a certain relation to the standard deviation of the observable:

For observables of the same sort, as distance measurements, or horizontal angles, or levelled height differences:

low value on the standard deviation	=>	large value of the weight
large value on the standard deviation	=>	low value of the weight

Exception:

If the observables in an estimation are of *different dimensions*, for instance distances and angles, the weights have different denominations (meter and gon), then the chosen denominations are deciding for the numerical values of the weights. See examples in the lecture notes.

3.1.1 Weight on the arithmetic mean

The arithmetic mean value of independent observations with the same accuracy of a variable, is, when the right hand side is separated into n fractions:

$$x = \frac{\sum l_i}{n} = \frac{l_1}{n} + \frac{l_2}{n} + \dots + \frac{l_n}{n} \quad (3.1)$$

Each of the n observation is contributing with $1/n$ to the mean value, x :

The mean value must therefore have n times larger weight than a single measurement, assuming the n observations have the same accuracy.

There are no conditions of the actual size of the weight *each observation* shall have, the weights are *relative figures*.

The weight of a single measurement: p

The weight of the arithmetic mean: p_x .

As the weights are relative, the weight of *one single observable* can be chosen. But after this choice the other weights are given / calculated.

1. We can choose that the weight of a single measurements in equation (3.1) shall have the weight 1:

$$p = 1.$$

The weight of the mean value is n times bigger than the weight of one single measurement. According to this the weight of the mean value is:

$$p_x = n \cdot 1 = n$$

2. We could also have chosen to let the mean value have the weight equal 1:

$$p_x = 1$$

A single measurement shall have a weight which is $1/n$ of the weight of the mean value:

$$p = p_x \cdot (1 / n) = 1 \cdot (1 / n) = 1 / n$$

This can also be seen from equation (3.1):

$$1 \cdot x = \frac{1}{n} l_1 + \frac{1}{n} l_2 + \cdots + \frac{1}{n} l_n$$

$$p_x \cdot x = p \cdot l_1 + p \cdot l_2 + \cdots + p \cdot l_n$$

In both the examples above dealing with choosing the observable with weight equal 1, the following relative condition is fulfilled:

$\frac{p_x}{p} = n$	<i>The weights are relative figures</i>	(3.2)
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3.1.2 Unit of weight. Measured value

Measured value

Measured value or observation are used as name of both a measured value from a single measurement and of derived measured values for instance from a calculation of a mean value of several measurements.

In a network of control points where the angles normally are measured in several rounds if a theodolite or a total station is used, the mean values of the rounds are used in the adjustments of the network. The mean values are considered as the measured values of the angles.

Unit of weight:

The measured value which we choose to give the weight $p = 1$, is named as the unit of weight. The unit of weight is normally a measured value (or we can also choose a derived measured value, like the mean value).

The standard deviation on the unit of weight is often named s_0 .

A measured value with the weight $p = 1$, has the standard deviation s_0 , or also named $\hat{\sigma}_0$

Examples of choice of unit of weight:

- Single measurement, see arithmetic mean in chapter 3.1.1
- The mean value of some of the measurements. See the example in chapter 5.4.2
- A height difference of a distance of 1 km. See the examples in chapter 5.5.1 and 5.5.2, (but not included here in 2010)

3.1.3 Variances and weights, the connection

The standard deviation of the arithmetic mean, s_x , is derived in chapter 5.2:

$$s_x = s_1 \cdot \frac{1}{\sqrt{n}}$$

, where s_1 is the standard deviation of a single measurement and n is the number of the measurements.

Obtaining the variance by squaring the standard deviation:

$$s_x^2 = s_1^2 \cdot \frac{1}{n} \quad \text{which is transformed to:} \quad \frac{s_1^2}{s_x^2} = n \quad (3.3)$$

Comparing the equations (3.2) and (3.3), and we obtain the following central connection between weight and variance:

$$\frac{s_1^2}{s_x^2} = \frac{p_x}{p_1} = n \quad \text{which is transformed to:} \quad s_1^2 \cdot p_1 = s_x^2 \cdot p_x \quad (3.4)$$

In equation (3.4) the connection between variance and weight is shown for arithmetic mean.

It can be shown that this connection is generally valid, and not only for arithmetic mean. There are n measured values with their own variances (squared standard deviation, s) and weights:

$$\text{Measured values:} \quad \mathbf{X} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \quad \text{Variances: } s_i^2 \quad \text{Weights: } p_i$$

NB: The variances are normally entered into a variance/covariance matrix, see (3.15), and the weights are entered into a weight matrix, see (3.10). Then the following relations are valid:

$$s_1^2 \cdot p_1 = s_2^2 \cdot p_2 = \dots = s_n^2 \cdot p_n \quad (3.5)$$

We choose a unit of weight with the standard deviation equal s_0 and with weight $p_0 = 1$. From equations (3.4) and (3.5):

$$s_i^2 \cdot p_i = s_0^2 \cdot 1 = \text{konstant} \quad (3.6)$$

The weight is inversely proportional to the variance.

Or:

The product of the variance and the weight of a parameter is a constant, and this constant is the variance of the unit of weight. .

This definition is valid generally and it is also valid for variables derived from observables or functions of these, and for observables of different dimensions (like distances and angles) when both these variables/observations are to be used in an adjustment.

Use of (3.6):

1. Known standard deviation, choose unit of weight. Can then calculate the weights from equation (3.6). *Example:* A measured distance with the length of 1 km has a standard deviation of $s_0 = 8$ mm, this observation is chosen as the unit of weight, $p = 1$. A distance of 2 km has the standard deviation of $s_2 = 12$ mm. The weight of the 2 km long distance is calculated to: $p_{2\text{km}} = (8 / 12)^2 = 0,444$

2. If the weights are known for a certain measuring method, the standard deviations can be calculated for the measurements done with this measuring method. See chapter 5.5 for weights on different methods.

3.2 Cofactors as accuracy measures

Earlier we have defined weight (p) as an accuracy measure. In several of the calculations where we want to find the standard deviation of derived variables, it is more convenient to use *co-variances* (q) instead.

The derivation of some of the fundamental connection between weight, co-factor and variance is first shown on stochastic independent variables, then the general case with correlated (dependent) variables.

3.2.1 Co-factors. Stochastic independent variables

Example: Horizontal directions measured by use of a total station.

In chapter 3.1.3 the following connection was deduced:

$$p_{xx} \cdot s_{xx}^2 = p_{yy} \cdot s_{yy}^2 = 1 \cdot s_{00}^2 \quad (3.7)$$

Where s_{00}^2 is the variance of the chosen unit of weight (the measurement with weight $p = 1$.)

Per definition the co-factor is the inverse of the weight, and we obtain for a variable (measurement):

$$q_{xx} = \frac{1}{p_{xx}} \quad \text{and} \quad q_{yy} = \frac{1}{p_{yy}} \quad (3.8)$$

Thus the co-factor is also to be regarded as an accuracy measure.

From the two equations (3.7) and (3.8) above and equation (3.13) later, we obtain the following important connection:

$$\sigma_{xx}^2 = \sigma_{00}^2 \cdot q_{xx} \quad \text{and} \quad \sigma_{yy}^2 = \sigma_{00}^2 \cdot q_{yy} \quad (3.9)$$

Using matrices:

The inverse of a weight matrix is called a *co-factor matrix* and is denoted $\mathbf{Q}$.

$$\mathbf{Q} = \mathbf{P}^{-1}$$

For *independent* observables the weight matrix is a diagonal matrix, the elements outside the main diagonal are equal to 0:

$$\mathbf{P}_{LL} = \begin{pmatrix} p_1 & 0 & \cdots & 0 \\ 0 & p_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & p_n \end{pmatrix} \quad (3.10)$$

The co-factor matrix of the observables is, according to the definition of co-factor:

$$\mathbf{Q}_{LL} = \mathbf{P}_{LL}^{-1} = \begin{pmatrix} \frac{1}{p_1} & 0 & \cdots & 0 \\ 0 & \frac{1}{p_2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \frac{1}{p_n} \end{pmatrix} = \begin{pmatrix} q_1 & 0 & \cdots & 0 \\ 0 & q_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & q_n \end{pmatrix} \quad (3.11)$$

In this case the $\mathbf{P}_{LL}$ matrix was easy to invert, as we have assumed uncorrelated observables and thus all the elements outside the main diagonal are equal to zero.

3.2.2 The co-factors. Stochastic correlated variables

Examples:

- Observations of horizontal angles in a station (see the example in chapter 5.1)
- A measured GPS baseline ($\Delta X, \Delta Y, \Delta Z$)
- Estimated coordinates.

In a co-factor matrix of correlated observables, the elements outside the main diagonal are normally different from 0. For instance the estimated coordinates are correlated, they have been “adjusted” to each other as there normally are more measurements than the unknowns.

The co-factor matrix with k variables:

$$\mathbf{Q} = \begin{pmatrix} q_{11}^2 & \cdots & q_{1k} \\ \vdots & \ddots & \vdots \\ q_{k1} & \cdots & q_{kk}^2 \end{pmatrix} \quad (3.12)$$

NB: The co-factor matrix is quadratic and symmetrical ($q_{ij} = q_{ji}$).

If some of the elements outside the main diagonal (q_{ij}) are different from zero, there is a correlation between the two actual variables.

In addition to equation (3.9) for the variance, the connection between the co-variance, the variance of the unit of weight and the co-factor is:

$$\sigma_{xy} = \sigma_{00}^2 \cdot q_{xy} \quad (3.13)$$

Example, a GPS baseline

The variance-covariance matrix of the measured baseline with the observations $\Delta X, \Delta Y$ and ΔZ in the geocentric coordinate system:

The co-factor matrix, $\mathbf{Q}_{XYZ}$, denomination m^2 , is shown in the table below. It is calculated as a part of the estimation of the coordinate differences. Notice that all the elements of the matrix are different from zero. You find the upper triangle of the co-factor matrix in the table, as the matrix is symmetrical around the main diagonal: *the three coordinate differences are correlated*.

a posteriori rms : 0.3957			
Co-factor matrix (upper triangle [m*m]) :			
	qx	qy	qz
qx	+5.9344000E-008	+5.1850000E-009	+5.4495001E-008
qy		+2.5848999E-008	+9.3720001E-009
qz			+2.2508399E-007

The variance-co-variance matrix of the estimated coordinate differences $\Delta X, \Delta Y$ and ΔZ , named $\mathbf{C}_{XYZ}$, is obtained from the known connection: $\mathbf{C}_{XYZ} = s_0^2 \cdot \mathbf{Q}_{XYZ}$, see equation (3.16).

Calculated value of the standard deviation of the unit of weight (*rms*) is $s_0 = 0,3957$, see the table.

The variance-co-variance matrix: $\mathbf{C}_{XYZ} = \frac{\begin{pmatrix} \sigma_{XX}^2 & \sigma_{XY} & \sigma_{XZ} \\ \sigma_{XY} & \sigma_{YY}^2 & \sigma_{YZ} \\ \sigma_{XZ} & \sigma_{YZ} & \sigma_{ZZ}^2 \end{pmatrix}}{\text{denomination mm}^2} = \begin{pmatrix} 0,00929 & 0,00081 & 0,00853 \\ 0,00081 & 0,00404 & 0,00147 \\ 0,00853 & 0,00147 & 0,03524 \end{pmatrix}$

The standard deviations on the estimated values of ΔX , ΔY and ΔZ of the baseline can be calculated from the variance-co-variance matrix by taking the square roots of the elements along the main diagonal.

Notice the size of the elements in the variance/covariance-matrix in the $\mathbf{C}_{XYZ}$ matrix above, they are too low (the accuracy is not that good...). It is recommended to scale the estimated standard deviation of the unit of weight by a factor of 8 to 10, to obtain more realistic values. More of this theme in other courses.

3.2.3 Calculation of the co-factor matrix, $\mathbf{Q}$

See chapter 4 about error propagation, also named variance/covariance propagation. In that chapter it is described how accuracy measures on derived variables can be calculated.

2010: Not included in this note: See chapter 10.2.16 for how to obtain the $\mathbf{Q}$ matrix of the estimated variables (for instance the coordinates and the heights), the matrix is calculated as a part of the estimation of the unknowns.

3.3 Variances and co-variances

We have defined weights, co-factors, standard deviation and variances as different measures of accuracy, and we have derived equations showing the connections between them.

When there are several variables (observations, derived variables) in an adjustment, there very often will be correlation between the variables.

The most easy-to-follow way to write the accuracy measures including the correlations is to use matrices.

In a variance/covariance matrix the variances are found along the main diagonal and the covariances outside the main diagonal. If a covariance element is different from zero, there is correlation between the two involved variables.

Below the process of defining the variance/covariance matrix is shown.

The stochastic variables (here two): $\mathbf{X} = \begin{pmatrix} x \\ y \end{pmatrix}$

Expected values of the variables: $E(\mathbf{X}) = \boldsymbol{\mu}_X = \begin{pmatrix} \mu_x \\ \mu_y \end{pmatrix}$

The true errors of the variables: $\boldsymbol{\varepsilon}_X = \begin{pmatrix} \varepsilon_x \\ \varepsilon_y \end{pmatrix} = \mathbf{X} - \boldsymbol{\mu} = E \begin{pmatrix} x - \mu_x \\ y - \mu_y \end{pmatrix}$

The variances (of x , similar for y): $\sigma_{xx}^2 = \text{Var}(x) = E[(x - \mu_x)^2] = E[\varepsilon_x \cdot \varepsilon_x]$

The covariance between x and y : $\sigma_{xy} = \text{Cov}(x, y) = E[(x - \mu_x)(y - \mu_y)] = E[\varepsilon_x \cdot \varepsilon_y]$

Now the variance-covariance matrix can be formed. We enlarge the number of observables to k . The variance/covariance matrix is found by multiplying: $\boldsymbol{\varepsilon}_X \cdot \boldsymbol{\varepsilon}_X^T = \mathbf{C}$

$$\mathbf{\varepsilon}_X \cdot \mathbf{\varepsilon}_X^T = \begin{pmatrix} \varepsilon_1 \cdot \varepsilon_1 & \cdots & \varepsilon_1 \cdot \varepsilon_k \\ \vdots & \ddots & \vdots \\ \varepsilon_k \cdot \varepsilon_1 & \cdots & \varepsilon_k \cdot \varepsilon_k \end{pmatrix} \quad (3.14)$$

See the definitions of the variance and the covariance above and compare with the elements of (3.14). The variance-covariance matrix, where the variances are along the main diagonal and the covariances are the elements outside the main diagonal:

$$\mathbf{C} = E[\mathbf{\varepsilon}_X \cdot \mathbf{\varepsilon}_X^T] = \begin{pmatrix} \sigma_{11}^2 & \cdots & \sigma_{1k} \\ \vdots & \ddots & \vdots \\ \sigma_{k1} & \cdots & \sigma_{kk}^2 \end{pmatrix} \quad (3.15)$$

- NB: The variance-covariance matrix is quadratic and symmetric ($\sigma_{ij} = \sigma_{ji}$).

Example:

The variables are the three estimated coordinates and height of the unknown point in a triangle, see the figure. If we assume that the coordinates x and y are estimated from horizontal angles and distances, and the heights H are calculated from the levelling, we obtain the following variance-covariance matrix:

$$\mathbf{X} = \begin{pmatrix} x \\ y \\ H \end{pmatrix} \quad \begin{array}{c} x, y, H \\ \triangle \end{array} \quad \mathbf{C}_{XX} = \begin{pmatrix} \sigma_{xx}^2 & \sigma_{xy} & 0 \\ \sigma_{xy} & \sigma_{yy}^2 & 0 \\ 0 & 0 & \sigma_{HH}^2 \end{pmatrix}$$

There are correlations between the two coordinates; while there are no correlations between the estimated height and the coordinates (two different measuring methods are used). This is shown in the variance/covariance matrix where the covariances between x and H and between y and H are equal zero. The covariance between x and y is different from zero, as the estimated coordinates are correlated.

3.4 Connection variances/covariances and co-factors

Comparing the equations (3.9) and (3.13), we can establish the following connections using matrices:

$$\mathbf{C} = \sigma_{00}^2 \cdot \mathbf{Q} \quad \text{or} \quad \mathbf{Q} = \frac{1}{\sigma_{00}^2} \cdot \mathbf{C} \quad (3.16)$$

Conclusions and comments:

- **Simulation:** The co-factor matrix is a relative variance-covariance matrix. The numerical values of the elements are dependent on the geometry and the measurements. By adding or removing one or more observations, the elements of co-factor matrix will change in numerical values. With that the co-factor matrix can be used in planning of which measurements we need to carry out to meet the requirements set for the results (for instance the quality of the estimated coordinates). We can *simulate* the network at office in advance and thus check the quality of the network.
- The denomination on the elements of the co-factor matrix can vary, dependent on the choice of the unit of weight.
- The variance of the unit of the weight is scaling the co-factor matrix to a variance-covariance matrix. In a simulation we can use an assumed value of the variance of the unit of weight.

4. The Gaussian error propagation law

Otherwise known as the *variance-covariance propagation law*.

Generally we measure GPS baselines, horizontal directions, horizontal angles, zenith angles and distances to be able to determine derived parameters such as coordinates, heights, areas etc. We also want to know the accuracy of the derived parameters, among other things to see if the accuracy measures fulfil the tolerances.

By using the error propagation law we calculate standard deviations and covariances of derived parameters, as a function of the standard deviations and covariances of the observables or from other parameters with a known variance-covariance matrix.

Two methods are used in this note:

1. *Error propagation law by using differentials*
2. *Error propagation law by using matrices*. General method, also for correlated observables.

4.1. Error propagation law by using differentials

In chapter 9 in the English text book this method is explained and used.

Normally we do not use this method, because the method using matrices is easier use, *especially when the parameters are correlated and if there are several parameters involved*.

In this course we emphasize the use of the “matrix method” which is explained in the next chapter.

The derivation of the error propagation law is not shown here, in the derivation mathematical methods for differentiations and series expansions are used.

The premise to use the law as it is developed in the 3 steps below: The observations are *independent variables*, as for instance measured horizontal directions or measured height differences are. For correlated observables, see point 4.

1. Mathematical connection between the observables, l_i , and the variable, x , we want to calculate the standard deviation on:

$$x = F(l_1, l_2, \dots, l_n)$$

2. The total differential is created by differentiation and series expansion (Taylor), where the series is cut after the 1st order terms.

$$x + dx = F(l_1, l_2, \dots, l_n) + \frac{\partial F}{\partial l_1} \cdot dl_1 + \dots + \frac{\partial F}{\partial l_n} \cdot dl_n \quad , \quad \frac{\partial F}{\partial l_i} = \alpha_i \quad (4.1)$$

$$dx = \alpha_1 \cdot dl_1 + \alpha_2 \cdot dl_2 + \dots + \alpha_n \cdot dl_n$$

3. We substitute the differential variables dx and dl with the standard deviations, s , squaring each term of the total differential, and thus obtaining the error propagation law.

NB: Assuming independent observables:

$$s_x^2 = \alpha_1^2 \cdot s_1^2 + \alpha_2^2 \cdot s_2^2 + \dots + \alpha_n^2 \cdot s_n^2 \quad (4.2)$$

4. If the observables are correlated, terms which contain the covariances must be added. An extended formulae for the error propagation law, with x_i as the correlated variables and y as the variable we want to find the standard deviation of:

$$y = F(x_1, x_2, \dots, x_n)$$

$$\sigma_y^2 = \left(\frac{\partial f}{\partial x_1} \right) \cdot \sigma_{x_1}^2 + \dots + \left(\frac{\partial f}{\partial x_n} \right) \cdot \sigma_{x_n}^2 + 2 \cdot \left(\frac{\partial f}{\partial x_1} \right) \cdot \left(\frac{\partial f}{\partial x_2} \right) \cdot \sigma_{x_1 x_2} + \dots + 2 \cdot \left(\frac{\partial f}{\partial x_{n-1}} \right) \cdot \left(\frac{\partial f}{\partial x_n} \right) \cdot \sigma_{x_{n-1} x_n} \quad (4.3)$$

For independent variables the covariances, $\sigma_{x_i x_j}$, are equal to zero. Equation (4.3) is then identical to equation (4.2).

4.2 Error propagation law by using matrices

The error propagation law using differentials in the previous chapter is Ok to use if we want to calculate the standard deviation of just one variable. If we want to find the accuracy measures in for instance a network of control points, using the matrix method is far more suitable. The matrix method is also much better to use when there are correlations between the variables.

Below the matrix method is derived, and the final result is the error propagation law on a general form and in an easy-to-use form. *The details of the derivation are not part of the curriculum in this course.*

The starting point is to find the mathematical connection between the observable, $\mathbf{X}$, and the parameter(s), $\mathbf{Y}$, we want to determine the standard deviation or the variance-covariance matrix of:

The mathematical connection is in general a function, $F(\mathbf{X})$: $\mathbf{Y} = F(\mathbf{X})$, where $\mathbf{X}$ contains the stochastic variables (for instance the observables), and we know the variance-covariance matrix of $\mathbf{X}$: $\mathbf{C}_{XX}$

The matrix $\mathbf{X}$ contains the stochastic variables, for instance the observables.

The problem is to find how $\mathbf{Y}$ behave (or to find the variance-covariance matrix of $\mathbf{Y}$), when we know the variance-covariance matrix of $\mathbf{X}$.

We must have *linear mathematical connections*:

- $\mathbf{Y} = f(\mathbf{X})$ is originally a linear function.
Examples: The function is a sum of observables (height differences from levelling, measurements using a measuring tape etc).
- $\mathbf{Y} = f(\mathbf{X})$ is originally not linear. We have to do a linearization.
Examples: Calculations of an area, coordinate estimations
The elements of the design matrix, $\mathbf{A}$, are determined by differentiation.

In the derivation of the law we assume linear connections, eventual non-linear connections are linearized. See chapter 4.3 (and 10.2.7, but not included here) for linearization

The original variables (for instance the observables), n variables: $\mathbf{X} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$

Transformed variables / parameters (for instance coordinates), e parameters: $\mathbf{Y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_e \end{pmatrix}$

Mathematical connection, transformation formula: $\mathbf{Y} = \mathbf{A} \cdot \mathbf{X} + \mathbf{B}$, where $\mathbf{A}$ and $\mathbf{B}$ are design matrices.

The connection between the **true values** of **Y** and **X** (the true value is what we seek but theoretically cannot find: What is the correct, true value of a distance or an angle?): True values, connection: $\mu_Y = A \cdot \mu_X + B$

True error is the difference between the observed (or calculated) value and the true value, the sign is opposite of the sign of the estimated correction, $\hat{v}$:

$$\begin{aligned}\epsilon_X &= X - \mu_X \\ \epsilon_Y &= Y - \mu_Y = A \cdot X - A \cdot \mu_X = A \cdot (X - \mu_X) \\ \epsilon_Y &= A \cdot \epsilon_X\end{aligned}$$

Using the formulas above and the definitions of the variances and the covariances (see chapter 3.3), we obtain the matrix formula below:

$$C_{YY} = E[\epsilon_Y \cdot \epsilon_Y^T] = E[A \cdot \epsilon_X \cdot \epsilon_X^T \cdot A^T]$$

As the **A** matrix is a design matrix (the elements are invariables), the expected value of the **A** matrix is the matrix itself:

$$E[A] = A$$

This leads to, as we can leave the **A** matrices outside the *E* function:

$$\begin{aligned}C_{YY} &= E[A \cdot \epsilon_X \cdot \epsilon_X^T \cdot A^T] = A \cdot E[\epsilon_X \cdot \epsilon_X^T] \cdot A^T \\ C_{YY} &= A \cdot C_{XX} \cdot A^T\end{aligned}\tag{4.4}$$

$$\text{From chapter 3.4: } C = \sigma_{00}^2 \cdot Q \quad \text{or} \quad Q = \frac{1}{\sigma_{00}^2} \cdot C\tag{4.5}$$

The error propagation law: Connecting (4.4) and (4.5) to:

$$Q_{YY} = A \cdot Q_{XX} \cdot A^T\tag{4.6}$$

Scaling the formula (4.6) with the variance of the unit weight, see (4.5), and we obtain the general *variance-covariance propagation law*. See also (4.4):

$$C_{YY} = A \cdot C_{XX} \cdot A^T\tag{4.7}$$

Remarks:

- The formulas (4.6) and (4.7) are generally valid, *also for correlated observables* (when the **Q_{XX}** or the **C_{XX}** matrix has values different from zero outside the main diagonal).
- The error propagation law is also valid if there are nonlinear connections between the variables in the **X** and the **Y** matrices.

Simulation:

We can use the error propagation law to analyze an actual planned measuring method in advance, to see which observables are most critical for the values of the accuracy measures of the parameter we want to determine.

This is called **simulation**: To find out in advance if the planned survey is Ok compared to the actual tolerances. See the examples!

Example: Linear mathematical connection:

Mathematical connection between y and x_i , a sum and/or a difference of several terms:

(originally linear connections): $y = a_1 \cdot x_1 + a_2 \cdot x_2 + \dots + a_n \cdot x_n$

We want to find the accuracy measures of the parameter y , when the accuracy measures of x_i are known. The elements of the design matrix are a_i , these elements are constants.

General equation using matrices, where $\mathbf{B} = 0$, as there are not any constants in equation above:

$$\mathbf{Y} = \mathbf{A}\mathbf{X} + \mathbf{B} = (y) = (a_1, a_2, \dots, a_n) \cdot \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} + 0$$

Assuming independent x parameters, the variance-covariance matrix of $\mathbf{X}$ will be a diagonal matrix (with only variances along the main diagonal and all covariances equal to zero):

$$\mathbf{C}_{XX} = \begin{pmatrix} \sigma_{11}^2 & 0 & \dots & 0 \\ 0 & \sigma_{22}^2 & & \\ \vdots & & \ddots & \vdots \\ 0 & & \dots & \sigma_{nn}^2 \end{pmatrix}$$

Variance-covariance propagation law, formula (4.7), $\mathbf{C}_{YY} = \mathbf{A} \cdot \mathbf{C}_{XX} \cdot \mathbf{A}^T$, where the $\mathbf{A}$ matrix is from the mathematical connection:

$$\mathbf{C}_{YY} = (\hat{\sigma}_{yy}^2) = (a_1 \quad a_2 \quad \dots \quad a_n) \cdot \begin{pmatrix} \sigma_{11}^2 & 0 & \dots & 0 \\ 0 & \sigma_{22}^2 & & \\ \vdots & & \ddots & \vdots \\ 0 & & \dots & \sigma_{nn}^2 \end{pmatrix} \cdot \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix}$$

Multiplying the matrices will give the accuracy measure of $\mathbf{Y}$, here y . As there is only 1 parameter in the $\mathbf{Y}$ matrix, we only have 1 element in the variance-covariance matrix of $\mathbf{Y}$: The variance of y .

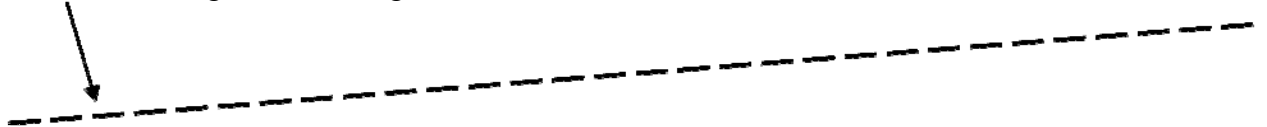
As the a_i coefficient is multiplied by itself, the sign of a_i ($\pm$) does not matter in the equation below (we always get +):

$$\hat{\sigma}_{yy}^2 = a_1^2 \cdot \hat{\sigma}_{11}^2 + a_2^2 \cdot \hat{\sigma}_{22}^2 + \dots + a_n^2 \cdot \hat{\sigma}_{nn}^2$$

This is the same result as by using the method of differentials on independent variables in chapter 4.1.

Example: Sum or differences of measurements**Here: Using prefabricated sections to build a bridge**

Pontoon bridge (floating bridge), assembled of prefabricated elements of the length of $l = 15,000$ m. The total length of the bridge: $D = 450,000$ m



- (1) The standard deviation of the length of one 15 meters long element is: $s_i = s = 4$ mm
Calculate the standard deviation of D , s_D .

Answer:

Mathematical connection between the elements and the total length:

There are $450 / 15 = 30$ elements: $D = l_1 + l_2 + \dots + l_{30}$

(a) Error propagation law by using differentials

$$dD = dl_1 + dl_2 + \dots + dl_{30}$$

$$s_D^2 = s_1^2 + s_2^2 + \dots + s_{30}^2$$

The standard deviations of the elements are s (the same for all): $s_x^2 = n \cdot s^2$

For numerical values see below, the matrix method is used.

(b) Mathematical connection using matrices: $\mathbf{D} = \mathbf{A}\mathbf{L} = (D) = (1 \ 1 \ \dots \ 1) \cdot \begin{pmatrix} l_1 \\ l_2 \\ \vdots \\ l_{30} \end{pmatrix}$

The measurement of each section is independent of the other sections: The covariances are zero. Error propagation law, $\mathbf{C}_{DD} = \mathbf{A} \cdot \mathbf{C}_{LL} \cdot \mathbf{A}^T$:

$$\mathbf{C}_{DD} = (\hat{\sigma}_D^2) = (1 \ 1 \ \dots \ 1)_{1 \times 30} \cdot \begin{pmatrix} s^2 & 0 & \dots & 0 \\ 0 & s^2 & & \\ \vdots & & \ddots & \vdots \\ 0 & & \dots & s^2 \end{pmatrix}_{30 \times 30} \cdot \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix}_{30 \times 1}$$

$$\begin{pmatrix} s^2 & 0 & \dots & 0 \\ 0 & s^2 & & \\ \vdots & & \ddots & \vdots \\ 0 & & \dots & s^2 \end{pmatrix} = s^2 \cdot \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & & \\ \vdots & & \ddots & \vdots \\ 0 & & \dots & 1 \end{pmatrix} = s^2 \cdot \mathbf{I}$$

$$\text{Multiplying: } \mathbf{C}_{DD} = (s_D^2) = (1 \ 1 \ \dots \ 1) \cdot s^2 \cdot \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix} = (30 \cdot s^2) \Rightarrow s_D^2 = 30 \cdot s^2$$

The standard deviation of D : $s_D = s \cdot \sqrt{30} = 22 \text{ mm}$

As the terms are added “in squares” the value of the standard deviation, 22 mm, is a much smaller value than obtained by a linear addition, $4 \cdot 30 = 120 \text{ mm}$. See chapter 4.4 about systematical errors.

(2) The standard deviation of D , s_D , calculated in (1) is too large.

What can be done to reduce the value of s_D ?

Answer:

This is an example of *simulation*, checking the accuracy in advance, before the production of the elements.

- Better production method of the elements, and thus reducing the standard deviation of the length of one element. If s_1 for instance is reduced to 2 mm (from 4 mm), the standard deviation of D is reduced to 11 mm (from 22 mm).
- Produce longer elements with the same standard deviation, 4 mm. If for instance the length is increased to 25 metres (18 elements instead of 30), the standard deviation of D is reduced to 17 mm ($4 \cdot \sqrt{18}$).

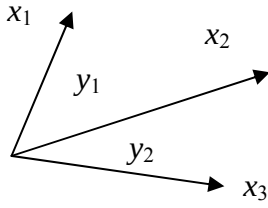
Example, horizontal directions and angles

The horizontal angles y_1 and y_2 are derived from the measured horizontal directions x_1 , x_2 and x_3 .

The horizontal directions are measured in a round (face left, face right etc).

The standard deviation of a measured direction for the actual total station (theodolite) is given as:

$s = 1,0$ mgon, all the directions are measured with the same precision, and the directions are generally considered as independent (not correlated).



The variance-covariance matrix of the directions is, from the given information's, equal to the unit matrix:

$$\mathbf{C}_{\mathbf{XX}} = \begin{pmatrix} 1,0^2 & 0 & 0 \\ 0 & 1,0^2 & 0 \\ 0 & 0 & 1,0^2 \end{pmatrix} = \mathbf{I}$$

The mathematical connection:

$$y_1 = x_2 - x_1 \quad \text{and} \quad y_2 = x_3 - x_2$$

using matrices: $\mathbf{Y} = \mathbf{AX} + \mathbf{B} = \begin{pmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$

The $\mathbf{A}$ matrix is used in the error propagation law, and we obtain the variance-covariance matrix of $\mathbf{Y}$:

$$\mathbf{C}_{\mathbf{YY}} = \mathbf{A} \cdot \mathbf{C}_{\mathbf{XX}} \cdot \mathbf{A}^T = \begin{pmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix} \cdot \mathbf{I} \cdot \begin{pmatrix} -1 & 0 \\ 1 & -1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$$

The standard deviations of the angles are: $s_{y1} = s_{y2} = \sqrt{2}$

The covariance is:

$$\sigma_{y1y2} = -1$$

The two angles are correlated, as the covariance is different from zero. The reason for this correlation is the direction x_2 which is common for these two angles. A negative covariance means in this case that an uncertainty in the measurement x_2 is affecting the two angles in different directions (increasing the value of one of the angles and decreasing the other angle). A value of -1 indicates a high degree of correlation.

4.3 Linearization of mathematical connections

In the previous pages and examples the mathematical connections between the stochastic variables and the parameter(s) we want to develop accuracy measures, was linear. In many cases this mathematical connection is not linear, examples are areas and coordinates.

The mathematical connection, a nonlinear function, generally written:

$$\mathbf{Y} = F(\mathbf{X}) = F(x_1, x_2, \dots, x_n)$$

The error propagation law is valid also for nonlinear functions, but we have to develop the elements in the design matrix $\mathbf{A}$ by partial derivations of the F function above, with respect of the different variables in the $\mathbf{X}$ matrix. Actually this method also works for cases with linear mathematical connections, but in this case we easily can pick the elements of the $\mathbf{A}$ matrix directly from the linear F function.

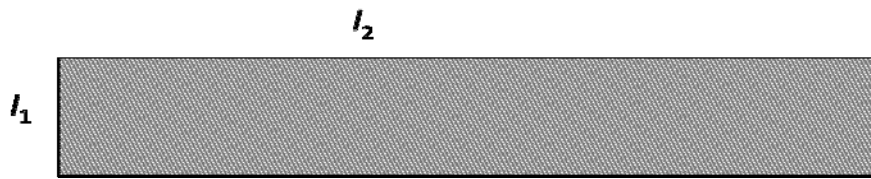
The $\mathbf{A}$ matrix in the error propagation law in the non-linear case:

$$\mathbf{A} = \begin{pmatrix} \frac{\partial F_1}{\partial x_1} & \dots & \frac{\partial F_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial F_r}{\partial x_1} & \dots & \frac{\partial F_r}{\partial x_n} \end{pmatrix} \quad (4.8)$$

The size of the $\mathbf{A}$ matrix is:

The number of variables in the $\mathbf{X}$ matrix is n : n columns

The number of F functions is r (r elements in the $\mathbf{Y}$ matrix): r rows

Example, nonlinear connections: The area, (y), of a rectangle

Observations: The two sides of the rectangle: $\mathbf{X} = \begin{pmatrix} l_1 \\ l_2 \end{pmatrix}$

The area of the rectangle: $\mathbf{Y} = (y) \Rightarrow 1 \text{ element}$

The dimension of the $\mathbf{A}$ matrix in this case is: $n = 2$ columns, $e = 1$ row

Originally mathematical connection: $y = F(l_1, l_2) = l_1 \cdot l_2$

The F function is nonlinear, we have to partial derivate to obtain the elements of the $\mathbf{A}$ matrix, a linearization using formula (4.8). The $\mathbf{A}$ matrix consists of 1 row with 2 elements and the $\mathbf{Y}$ matrix has only 1 element (one function, the area):

$$\mathbf{A} = (l_2 \cdot 1, l_1 \cdot 1) = (l_2, l_1)$$

The error propagation law where we assume the two distance measurements to be independent of each other (the covariance is zero):

$$\mathbf{C}_{\mathbf{Y}\mathbf{Y}} = (s_{yy}^2) = (l_2 \quad l_1) \cdot \begin{pmatrix} s_1^2 & 0 \\ 0 & s_2^2 \end{pmatrix} \cdot \begin{pmatrix} l_2 \\ l_1 \end{pmatrix}$$

Multiplying: $s_y^2 = l_2^2 \cdot s_1^2 + l_1^2 \cdot s_2^2$

Numerical values and simulation:

$$l_1 = 10,00 \text{ m} \quad s_1 = 0,2 \text{ m (measured on a map?)}$$

$$l_2 = 100,00 \text{ m} \quad s_2 = 0,2 \text{ m (measured on a map?)}$$

The area: $y = 100 \cdot 10 = 1000 \text{ m}^2$

The variance of the area: $s_{yy}^2 = l_2^2 \cdot s_1^2 + l_1^2 \cdot s_2^2 = (100 \cdot 0,2)^2 + (10 \cdot 0,2)^2 = 400 + 4 = 404 \text{ m}^4$

The standard deviation of the area: $s_{yy} = \sqrt{404} = 20,1 \text{ m}^2$

Simulation:

Notice we can calculate the variance-covariance matrix without using actual / real measurements. We just need information of the geometry (here a rectangle) and the variance-covariance matrix of the variables (here the two sides). This means we can find out in advance if the measuring method and the choice of the instruments are Ok, by using the error propagation law to check if the result is Ok.

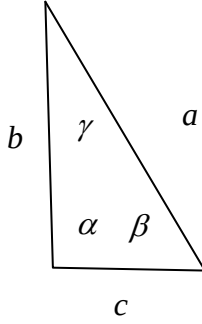
Assume the calculated standard deviation of 20,1 is larger than the tolerance: Which side of the rectangle is it most important to measure with the highest accuracy, if the aim is to reduce the standard deviation of the area, y ? From the figures in the variance formula: The (in)accuracy of l_1 (the shortest side) is contributing with 400 of the 404. $\Rightarrow l_1$ is critical.

If we measure l_1 with a standard deviation of 0,02 m (by using a total station) instead of 0,2 m, the variance on the area is: $s_{yy}^2 = l_2^2 \cdot s_1^2 + l_1^2 \cdot s_2^2 = (100 \cdot 0,02)^2 + (10 \cdot 0,2)^2 = 4 + 4 = 8 \text{ m}^4$

The standard deviation of the area: $s_{yy} = \sqrt{8} = 2,8 \text{ m}^2$

Example: Measurements in a triangle

Two angles, α and β , and one side, c , in a triangle are measured, see the figure.



The distance a is not possible to measure, but this distance is critical for the project:

The standard deviation of a is to be calculated.

Mathematical connections are not originally linear, as trigonometrical functions are involved. The sine proportion:

$$a = F(\alpha, \beta, c) \Rightarrow a = \frac{c \cdot \sin \alpha}{\sin(\gamma)} = \frac{c \cdot \sin \alpha}{\sin(200 - \alpha - \beta)} = \frac{c \cdot \sin \alpha}{\sin(\alpha + \beta)}$$

Using formula (4.8) to create the **A** matrix, we have three variables in this case: α , β and c . The 3 elements of the **A** matrix are:

$$\frac{\partial F}{\partial c} = \frac{a}{c} \qquad \frac{\partial F}{\partial \alpha} = \frac{c \cdot \sin \beta}{\sin^2 \gamma} \qquad \frac{\partial F}{\partial \beta} = -\frac{a}{\tan \gamma}$$

We assume the three observables are independent. By using the error propagation law, formula (4.7) we obtain:

$$s_a^2 = \left(\frac{a}{c}\right)^2 \cdot s_c^2 + \left(\frac{c \cdot \sin \beta}{\sin^2 \gamma}\right)^2 \cdot \frac{s_\alpha^2}{\rho^2} + \left(\frac{a}{\tan \gamma}\right)^2 \cdot \frac{s_\beta^2}{\rho^2} \quad , \rho = 200000 \text{ mgon} / \pi$$

Inserting numerical values:

$$\begin{aligned} c &= 200,000 \text{ m} & s_c &= 8,5 \text{ mm} \\ \alpha &= 100,1112 \text{ gon} & s_\alpha &= 2,0 \text{ mgon} \\ \beta &= 66,6667 \text{ gon} & s_\beta &= 1,0 \text{ mgon} \\ a &\text{ is calculated to be } 401,214 \text{ m.} \end{aligned}$$

The answer is in metre or millimetre. The standard deviations of the angles must be given in radians (to "get rid of" the denomination mgon in the equation). Therefore the conversion factor ρ is introduced, here going from mgon to radians. Also important to be consistent and use the same denomination for the measures of lengths. Here metre is used as denomination of distances and standard deviations of the distances:

$$s_a^2 = \left(\frac{401,214}{200,000}\right)^2 \cdot 0,0085^2 + \left(\frac{200,000 \cdot \sin 66,6667}{\sin^2 33,2221}\right)^2 \cdot \frac{2,0^2}{63662^2} + \left(\frac{401,214}{\tan 33,2221}\right)^2 \cdot \frac{1,0^2}{63662^2} \quad (4.9)$$

$$s_a^2 = (289 + 481 + 118) \cdot 10^{-6} \text{ m}^2$$

$$s_a = \sqrt{888} = 29,8 \text{ mm}$$

See also the next chapter for more use of this example.

4.4 The error propagation law: Adding in squares. Simulation

The last example in the last chapter (the triangle) is here used to stress two important characteristics of the error propagation law:

- Notice in equation (4.9) the three terms are *added in squares*: $s_a^2 = S_c^2 + S_\alpha^2 + S_\beta^2$

The error propagation of random errors of the measurements has a favourable accumulation. The random errors vary in size and *sign* (compare with the clock curve of the normal distribution). As a consequence of this, the total standard deviation in (4.9) has a lower value

than we obtain using a linear addition. See also the comparison with the systematical error in chapter 4.5, figure 4.3.

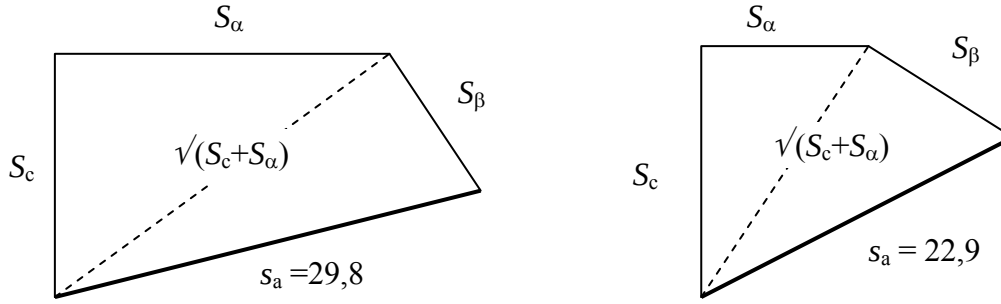


Figure 4.2 Random errors: Adding in squares

- **Simulation:** The calculations of the numerical values of a standard deviation of a variable can be performed before doing any observations in the field. Assumed standard deviations are inserted on the left hand side in the error propagation law, equation (4.6). The geometry and the planned measurements give the mathematical connection which later is to be used in the error propagation law.

In equation (4.9), the variance of the distance a , the term in the middle, has a value of 481, which is more than the sum of other two terms. If we want to reduce the standard deviation of a , it is most efficient to find a way to reduce the numerical value of this term, the contribution from the measured angle α . If we measure the angle α with the same accuracy as the angle β , thus will reduce the standard deviation to $s_\alpha = 1,0$ mgon. Method: Measure more rounds or use a theodolite (total station) with a lower standard deviation. With $s_\alpha = 1,0$ mgon instead of $s_\alpha = 2,0$ mgon, the term is reduced to 1/4th of the value in equation (4.9):

$$s_a^2 = (289 + 120 + 118) \cdot 10^{-6} \text{ m}^2 \quad \Rightarrow \quad s_a = \sqrt{527} = 22,9 \text{ mm} \quad (4.10)$$

In (4.10) we observe it is the accuracy of the distance c (the first term) that has the largest contribution to the value of the standard deviation of a .

In a simulation of a network of control points we can add new measurements until the wanted accuracy measures are achieved in all the points in the network. We can also consider if another observation technique or using other (and better) instruments will give better accuracies of the results.

4.5 Systematical errors and error propagation

We have during the development and the use of the error propagation law assumed we only are dealing with random errors. But, there will also be observations that are more or less influenced of systematical errors, and systematical errors do not follow the Gaussian error propagation law.

Below is an investigation of how the systematical errors on the observables will influence a function of the same observables.

4.5.1 Derivation of formulas

According to (4.1) the true error of the function x , is derived from the true errors of the observations l_i :

$$\varepsilon_x = \sum (\alpha_i \cdot \varepsilon_i) \quad \text{where} \quad \alpha_i = \frac{\partial F}{\partial l_i} \quad (4.11)$$

When systematic errors occur, the total true error, ε , can be split into a random error, δ , and a systematic error, η : $\varepsilon_i = \delta_i + \eta_i$

$$\varepsilon_i = \delta_i + \eta_i \text{ is inserted into equation (4.11): } \varepsilon_x = \sum \alpha_i \cdot (\delta_i + \eta_i) \quad (4.12)$$

The observation series which gives (4.12), are performed several times, ν (ny). For all the series (4.12) are squared and divided by ν .

According to the definition of the standard deviation, equation (2.1), we obtain:

$$\hat{\sigma}_x^2 = \frac{\sum \varepsilon_x^2}{\nu} = \sum \alpha_i^2 \sigma_i^2 + \sum \alpha_i^2 \eta_i^2 + \sum_{i=1, j=i+1}^{i=n-1, j=n} 2\alpha_i \alpha_j \eta_i \eta_j \quad (4.13)$$

In (4.13) are the Σ terms that converge to zero omitted (see equivalent considerations after equation (2.13)). There are terms containing the random error, δ_i , as a factor that are left out.

4.5.2 Error propagation

Equation (4.13) arranged in another way:

$$\hat{\sigma}_x^2 = \sum \alpha_i^2 \sigma_i^2 + (\alpha_1 \eta_1 + \alpha_2 \eta_2 + \dots + \alpha_n \eta_n)^2 \quad (4.14)$$

The Σ term in (4.14) contains the influence of the random errors following the Gaussian error propagation law. The last term is the influence of the systematic errors, η .

A simplified case of equation (4.14) is used below. We assume the observables to have the same standard deviations and the same systematic error. Example: Using a measuring tape.

Then: $\alpha_i = 1$ $\sigma_i = s_i = s$ and $\eta_i = \eta$ for all the i

$$\hat{s}_x^2 = \hat{\sigma}_x^2 = n \cdot s^2 + n^2 \cdot \eta^2 \quad (4.15)$$

Equation (4.15) shows that a systematic error has much stronger error propagation than the random errors, see the factors n and n^2 in the terms of respectively random and systematic errors. See figure 4.3 in the example in chapter 4.5.3 below.

4.5.3 Example: Co-operation between random and systematic errors

A distance is measured by using a 25 m long measuring tape. The tape is calibrated on a control basis. The calibration gave a standard deviation of 1,6 mm on a distance of 25 m. Therefore we have to assume that the tape has a remaining systematic error which could be either positive or negative, and this error can only be given as $s_k = 1,6$ mm. This is an uncertainty of the tape which is of systematic character, as the true value of the tape can differ from the measured value of 25,000 m.

The distance between 0 and 25 metre mark on the tape is found to be: 25,000 m $\pm$ 1,6 mm. In addition to this systematic error above we have the random errors occurring when we use the tape: The placing of the tape in both ends, the readings on the tape, random variations of the tensions, random oscillations in temperature etc. For each measured length of the tape we assume a standard deviation caused by these random errors to be: $s_1 = 3,2$ mm, which is ca 2 times larger than the systematic error of 1,6 mm.

A distance of $L = 200$ m is measured with this tape.

Inserted into equation (4.15), with $n = 200 / 25 = 8$, we obtain:

$$s_L = \sqrt{8 \cdot 3,2^2 + 8^2 \cdot 1,6^2} = \sqrt{82 + 164} = 15,7 \text{ mm}$$

The systematic error is half the size of the standard deviation of the random error, but the contribution from the systematic error to the standard deviation of L , is twice as large as the contribution from the random errors.

Figure 4.3 shows the co-operation between systematic error and the random errors and how the influence of the systematic error is more and more dominating with increasing values of n (the number of observations).

*The systematic errors have linear accumulations.
The standard deviations of random errors are added in squares.*

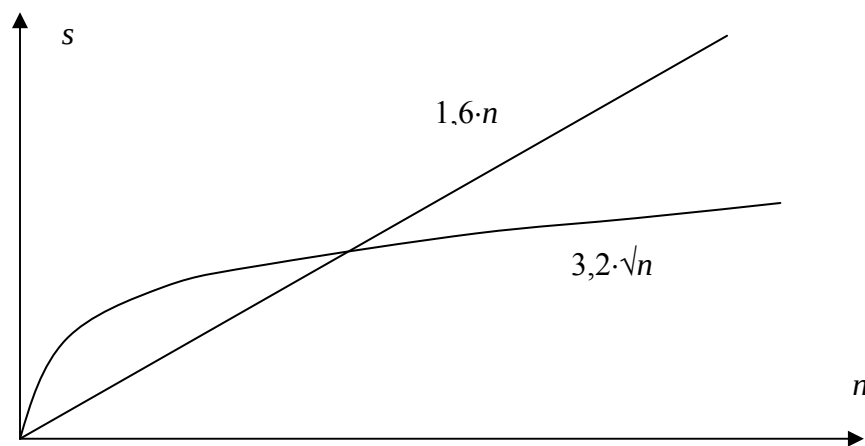


Figure 4.3 Error propagation of random and systematic errors with increasing value of n

5. Applications of the error propagation law

Only the method using matrices is used here. See in your main textbook for similar calculations where the method based on differentials is used.

5.1 Sum or differences of stochastic independent observables

Mathematical connection (originally linear), where y can be partly a sum or a difference of observables, x_i . The coefficients, a_i , are in this case either $+1$ or -1 :

$$y = a_1 \cdot x_1 + a_2 \cdot x_2 + \dots + a_n \cdot x_n$$

$$y = \pm 1 \cdot x_1 \pm 1 \cdot x_2 \pm \dots \pm 1 \cdot x_n$$

General equations by using matrices:

$$\mathbf{Y} = \mathbf{A}\mathbf{X} + \mathbf{B} = (y) = (\pm 1, \pm 1, \dots, \pm 1) \cdot \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} + 0$$

Independent observables, x , giving the following variance-covariance matrix of $\mathbf{X}$, all the covariances outside the main diagonal are zero:

$$\mathbf{C}_{xx} = \begin{pmatrix} \hat{\sigma}_{11}^2 & 0 & 0 & 0 \\ 0 & \hat{\sigma}_{22}^2 & 0 & 0 \\ 0 & 0 & \ddots & 0 \\ 0 & 0 & 0 & \hat{\sigma}_{nn}^2 \end{pmatrix}$$

The error propagation law, using equation (4.7):

$$\mathbf{C}_{\mathbf{Y}\mathbf{Y}} = (\hat{\sigma}_{yy}^2) = (\pm 1 \quad \pm 1 \quad \dots \quad \pm 1) \cdot \begin{pmatrix} \hat{\sigma}_{11}^2 & 0 & 0 & 0 \\ 0 & \hat{\sigma}_{22}^2 & 0 & 0 \\ 0 & 0 & \ddots & 0 \\ 0 & 0 & 0 & \hat{\sigma}_{nn}^2 \end{pmatrix} \cdot \begin{pmatrix} \pm 1 \\ \pm 1 \\ \vdots \\ \pm 1 \end{pmatrix}$$

Multiplying, where the coefficients are squared and thus the signs of these coefficients, +1 and -1, have nothing to say (will always be +1), and we obtain:

$$\hat{\sigma}_{yy}^2 = \hat{\sigma}_{11}^2 + \hat{\sigma}_{22}^2 + \dots + \hat{\sigma}_{nn}^2 \quad (5.1)$$

If all the observables are of the same quality, they have the same standard deviation. Here two notations of the empirical standard deviations are used:

$$s_1 = s_2 = \dots = s_n \quad \text{or} \quad \hat{\sigma}_1 = \hat{\sigma}_2 = \dots = \hat{\sigma}_n$$

Inserted into (5.1):

$$\begin{aligned} \hat{\sigma}_{yy}^2 &= \hat{\sigma}^2 + \hat{\sigma}^2 + \dots + \hat{\sigma}^2 \\ \hat{\sigma}_{yy}^2 &= n \cdot \hat{\sigma}^2 \end{aligned} \quad (5.2)$$

For the standard deviations we have in this case, with observables of the same quality:

$$\boxed{\hat{\sigma}_{yy} = \hat{\sigma} \cdot \sqrt{n}} \quad (5.3)$$

The standard deviation of a sum (or difference) of independent observables of the same quality are proportional with the square root of the number of the measurements that are involved in obtaining the sum (difference).

Example: Horizontal angle, β , derived from two directions, x_2 and x_1 :

The precision of a theodolite (total station) is normally given by the standard deviation of a reading of a horizontal direction, s_r .

Standard deviation of a horizontal angle, s_β , can be derived from s_r by using the error propagation law:

Mathematical connection: $\beta = r_2 - r_1$

Mathematical connection using matrices: $\mathbf{Y} = \mathbf{A}\mathbf{X} = (\beta) = (-1 \quad +1) \cdot \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$

$\Rightarrow$ The standard deviation of the angle, see also the example at the end of chapter 4.2:

$$s_\beta = s_r \cdot \sqrt{2}$$

Example: Sum of measurements

$$y = x_1 + x_2 + x_3 \quad \Rightarrow \quad \mathbf{Y} = \mathbf{A}\mathbf{X} = (y) = (1 \quad 1 \quad 1) \cdot \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

The standard deviation of the sum: $\hat{\sigma}_y = \hat{\sigma}_x \cdot \sqrt{3}$

Example with numerical values: Measuring of a distance

A distance of 200 metres in an industry plant is measured with a measuring tape which is 25 m long.

The standard deviation of one measurements of a distance of 25 m, is found to be $s = 3$ mm.

The standard deviation of 200 m by using this tape is calculated by using equation (5.3), $n = 200 / 25 = 8$ measurements of 25 m:

$$s_{200} = \hat{\sigma}_{200} = 3 \cdot \sqrt{8} = 8,5 \text{ mm}$$

5.2 The standard deviation of arithmetical mean value

The arithmetical mean value is obtained by the following mathematical connection, see chapter 2.3. It is a linear mathematical connection:

$$\hat{x} = \frac{\sum l_i}{n} = \frac{1}{n} \cdot l_1 + \frac{1}{n} \cdot l_2 + \dots + \frac{1}{n} \cdot l_n$$

Using matrices: $\mathbf{X} = \mathbf{A}\mathbf{L} + \mathbf{B} = (\hat{x}) = \begin{pmatrix} \frac{1}{n} & \frac{1}{n} & \dots & \frac{1}{n} \end{pmatrix} \cdot \begin{pmatrix} l_1 \\ l_2 \\ \vdots \\ l_n \end{pmatrix}$

Independent observables, l , gives the following variance-covariance matrix of $\mathbf{L}$, the covariances outside the main diagonal are all equal to zero:

$$\mathbf{C}_{\mathbf{LL}} = \begin{pmatrix} \hat{\sigma}_{11}^2 & 0 & 0 & 0 \\ 0 & \hat{\sigma}_{22}^2 & 0 & 0 \\ 0 & 0 & \ddots & 0 \\ 0 & 0 & 0 & \hat{\sigma}_{nn}^2 \end{pmatrix}$$

The error propagation law, using equation (4.7):

$$\mathbf{C}_{\mathbf{xx}} = (\hat{\sigma}_{xx}^2) = \begin{pmatrix} \frac{1}{n} & \frac{1}{n} & \dots & \frac{1}{n} \end{pmatrix} \cdot \begin{pmatrix} \hat{\sigma}_{11}^2 & 0 & 0 & 0 \\ 0 & \hat{\sigma}_{22}^2 & 0 & 0 \\ 0 & 0 & \ddots & 0 \\ 0 & 0 & 0 & \hat{\sigma}_{nn}^2 \end{pmatrix} \cdot \begin{pmatrix} \frac{1}{n} \\ \frac{1}{n} \\ \vdots \\ \frac{1}{n} \end{pmatrix}$$

Multiplying the matrices: $\hat{\sigma}_{xx}^2 = \frac{1}{n^2} \cdot (\hat{\sigma}_{11}^2 + \hat{\sigma}_{22}^2 + \dots + \hat{\sigma}_{nn}^2)$ (5.4)

If all the observables are of the same quality, they have the same standard deviation:

$$s = s_1 = s_2 = \dots = s_n \quad \text{or} \quad \hat{\sigma} = \hat{\sigma}_1 = \hat{\sigma}_2 = \dots = \hat{\sigma}_n$$

Inserted into (5.4): $\hat{\sigma}_{xx}^2 = \frac{1}{n^2} (\hat{\sigma}^2 + \hat{\sigma}^2 + \dots + \hat{\sigma}^2)$, $\hat{\sigma}_{xx}^2 = \frac{1}{n} \cdot \hat{\sigma}^2$ (5.5)

For the standard deviations we have in this case, with observables of the same quality:

$$\hat{\sigma}_{xx} = \hat{\sigma} \cdot \frac{1}{\sqrt{n}}$$

(5.6)

The standard deviation of arithmetical mean value made of independent observables of the same quality, is inverse proportional with the square root of the number of observations that are involved in obtaining the arithmetical mean value.

Example with numerical values: Measuring of a horizontal angle.

A horizontal angle is measured 6 times, the measurements are independent of each other and are of the same quality (same weight). The observations, arithmetical mean value and estimated corrections are shown in the table below:

Observations, l , gon	Estimated corrections, $\hat{v} = x - l$, mgon	$\hat{v}^2$
50,4121	+0,6	0,36
50,4124	+0,3	0,09
50,4127	0,0	0,00
50,4135	-0,8	0,64
50,4125	+0,2	0,04
50,4130	-0,3	0,09
$\hat{x} = (\sum l)/n = 50,4127$	$\sum v = 0,0$	$\sum v^2 = 1,22$

The standard deviation of one measurement, equation (2.1): $\hat{s}_0 = \sqrt{\frac{\sum \hat{v}^2}{n-1}} = \sqrt{\frac{1,22}{5}} = 0,49$ mgon

The standard deviation of the arithmetical mean value, equation (5.6):

$$\hat{s}_x = \frac{s_0}{\sqrt{n}} = \frac{0,49}{\sqrt{6}} = 0,20 \text{ mgon}$$

5.2.1 The law of diminishing utility of increasing number of measurements

We assume the standard deviation of one measurement is known. Empirically we have good values of this standard deviation, as the technical data for a specific total station is normally given from the producer, or we can carry out our own accuracy investigations (by measuring the same angle a vast number of times).

The observable is an angle measured 1 time, with a standard deviation of $\hat{\sigma}$. By applying equation (5.6) we see that if we measure the actual parameter 4 times (here: measure the angle in 4 rounds), the standard deviation of the arithmetical mean value of the 4 measurements:

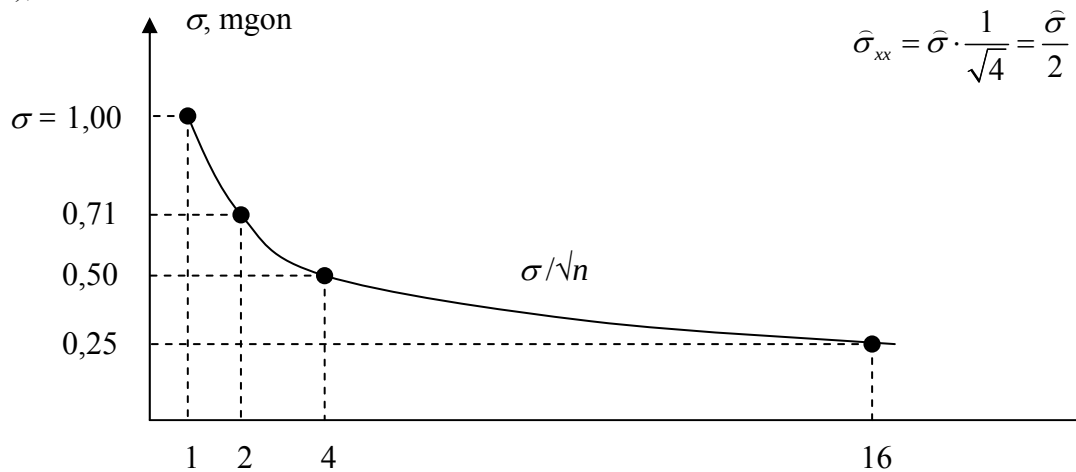


Figure 5.1 The law of diminishing utility. Angle measurements

As we can see from the figure, the standard deviation of the mean value is halved by increasing the measurements from 1 to 4. A further halving of the standard deviation is obtained by increasing the measurements from 4 to 16.

This example is illustrating **the law of diminishing utility of increasing number of measurements**. We obtain a lot by increasing the number of measurements from 1 to 2, 3 or 4, but a further increasing is somewhat a waste of time, compared to what we achieve of better accuracy measures. See also figure 5.1.

5.3 Weighted mean. Observables of various accuracy

In chapter 5.2 the topic was *arithmetical mean value*, where all the measurements which contribute to the mean value have the same accuracy, the same weight.

In this chapter the general case with observables of different weights is presented.

The mean value in this case is called *weighted mean value*.

5.3.1 Calculation of weighted mean

Assuming a parameter, for instance a horizontal angle, is measured 12 times in a quality check of a total station. The measurements are carried out in three stages:

Stage 1: 4 rounds, arithmetical mean value is x_1

Stage 2: 5 rounds, arithmetical mean value is x_2

Stage 3: 3 rounds, arithmetical mean value is x_3

If all the 12 measurements are carried out with the same accuracy, they have the same standard deviation, s . Using equation (5.6) we obtain: $s_{x1}^2 = \frac{s^2}{4}$ $s_{x2}^2 = \frac{s^2}{5}$ $s_{x3}^2 = \frac{s^2}{3}$

In calculating the final mean value of x_1 , x_2 and x_3 , it is logical to weight these three mean values by using the number of rounds. As the observable of unit weight we choose a single measurement with the standard deviation s .

x_1 with the weight $p_1 = 4$ x_2 with the weight $p_2 = 5$ x_3 with the weight $p_3 = 3$

These three weights are fulfilling the demand in equation (3.6), that the product of the variance and the weight must be constant, in this case equal to s^2 .

The weighted mean value is:

$$\hat{x} = \frac{\sum x_i \cdot p_i}{\sum p_i} = \frac{x_1 \cdot 4 + x_2 \cdot 5 + x_3 \cdot 3}{4 + 5 + 3} \quad (5.7)$$

Equation (5.7) gives of course the same mean value as if the mean value is calculated directly from the original 12 single measurements. This is easily seen from (5.7), when x_1 , x_2 and x_3 are substituted with the observations they are calculated from:

$$x_1 = \frac{\sum_{i=1}^4 l_i}{4} \quad x_2 = \frac{\sum_{i=1}^9 l_i}{5} \quad x_3 = \frac{\sum_{i=1}^{12} l_i}{3}$$

Which gives: $x = \frac{\sum_{i=1}^{12} l_i}{12}$, the arithmetical mean where the 12 measurements have the same weight, $p = 1$

General formula for the weighted mean value:

$$\boxed{\hat{x} = \frac{\sum_{i=1}^n l_i \cdot p_i}{\sum_{i=1}^n p_i}} \quad (5.8)$$

5.3.2 The standard deviation of the weighted mean

The standard deviation of the weighted mean in equation (5.8) is found by applying the error propagation law on the mathematical connection in (5.8).

Using matrices: $\mathbf{X} = \mathbf{A}\mathbf{L} = (\hat{x}) = \frac{1}{\sum p} \cdot (p_1 \ p_2 \ \cdots \ p_n) \cdot \begin{pmatrix} l_1 \\ l_2 \\ \vdots \\ l_n \end{pmatrix}$

Independent observables, l , gives the following variance-covariance matrix of $\mathbf{L}$, the covariances outside the main diagonal are all equal to zero:

$$\mathbf{C}_{\mathbf{L}\mathbf{L}} = \begin{pmatrix} \hat{\sigma}_{11}^2 & 0 & 0 & 0 \\ 0 & \hat{\sigma}_{22}^2 & 0 & 0 \\ 0 & 0 & \ddots & 0 \\ 0 & 0 & 0 & \hat{\sigma}_{nn}^2 \end{pmatrix}$$

The error propagation law, using equation (4.7):

$$\mathbf{C}_{\mathbf{X}\mathbf{X}} = (\hat{\sigma}_{xx}^2) = \frac{1}{\sum p} \cdot (p_1 \ p_2 \ \cdots \ p_n) \cdot \begin{pmatrix} \hat{\sigma}_{11}^2 & 0 & 0 & 0 \\ 0 & \hat{\sigma}_{22}^2 & 0 & 0 \\ 0 & 0 & \ddots & 0 \\ 0 & 0 & 0 & \hat{\sigma}_{nn}^2 \end{pmatrix} \cdot \begin{pmatrix} p_1 \\ p_2 \\ \vdots \\ p_n \end{pmatrix} \cdot \frac{1}{\sum p}$$

Multiplying the matrices:

$$\hat{\sigma}_{xx}^2 = \frac{1}{(\sum p)^2} \cdot (p_1^2 \cdot \hat{\sigma}_{11}^2 + p_2^2 \cdot \hat{\sigma}_{22}^2 + \cdots + p_n^2 \cdot \hat{\sigma}_{nn}^2) \quad (5.9)$$

According to (3.6), giving the connection between the variance and the weight, where $\hat{\sigma}_0$ (also written as $\hat{\sigma}_{00}$) is the standard deviation of the observable we have chosen to be of unit weight, $p = 1$:

$$p_i \cdot \hat{\sigma}_i^2 = 1 \cdot \hat{\sigma}_0^2$$

Inserted in (5.9):

$$\begin{aligned} \hat{\sigma}_{xx}^2 &= \frac{1}{(\sum p)^2} \cdot (p_1 \cdot \hat{\sigma}_{00}^2 + p_2 \cdot \hat{\sigma}_{00}^2 + \cdots + p_n \cdot \hat{\sigma}_{00}^2) \\ \hat{\sigma}_{xx}^2 &= \frac{\hat{\sigma}_{00}^2}{\sum p_i} \end{aligned} \quad (5.10)$$

See chapter 5.4 for the standard deviation of the observable with unit weight when weighted mean is used, $\hat{\sigma}_0$

For the standard deviations we have in this case, with observables of the different quality:

$$\boxed{\hat{\sigma}_{xx} = \frac{\hat{\sigma}_{00}}{\sqrt{\sum p_i}}} \quad (5.11)$$

The standard deviation of the weighted mean calculated from independent observables of different weights, is inverse proportional to the square root of the sum of the weights of the observables that are involved in obtaining the weighted mean

If all the measurements have the same weight, $p = 1$, formula (5.11) will give the standard deviation of the arithmetical mean value:

$$\hat{\sigma}_{xx} = \frac{\hat{\sigma}_{00}}{\sqrt{n}}$$

5.3.2 The weight of weighted mean value

From (5.11): $\hat{\sigma}_{xx}^2 \cdot \sum p_i = \hat{\sigma}_{00}^2$ From (3.6): $\hat{\sigma}_{xx}^2 \cdot p_x = 1 \cdot \hat{\sigma}_{00}^2$

Comparing the two formulas above, and we obtain the weight of the weighted mean:

$$\boxed{p_x = \sum p_i} \quad (5.12)$$

The weight of the mean value of observables with different weights (weighted mean) is equal to the sum of the weights of the observables.

5.4 The standard deviation of the unit weight, observables of different weights

We have in chapter 2.3 derived the formula of the standard deviation of the unit weight, see arithmetical mean in chapter 2.3. Here the same procedure is used to find the standard deviation of the unit weight, when the observables are of different weight. There are other and perhaps more elegant ways (?) to find the formula, see literature in mathematical statistics.

The observations (assumed to follow the normal distribution or to be approximately normal distributed) and the accompanying weights along the main diagonal in the weight matrix:

$$\mathbf{L} = \begin{pmatrix} l_1 \\ l_2 \\ \vdots \\ l_n \end{pmatrix} \quad \mathbf{P} = \begin{pmatrix} p_1 & & & \\ & p_2 & & \\ & & \ddots & \\ & & & p_n \end{pmatrix}$$

The true value of the mean value is μ . The weighted mean value, $\bar{x}$, is an estimator of μ :

$$\bar{x} = \frac{\sum_{i=1}^n l_i \cdot p_i}{\sum_{i=1}^n p_i}$$

The goal is to show that the standard deviation of the weighted mean is:

$$\boxed{\hat{\sigma}_0 = \sqrt{\frac{\sum p \hat{\varepsilon} \hat{\varepsilon}}{n-1}} = \sqrt{\frac{\sum p \hat{v} \hat{v}}{n-1}}} \quad (5.13)$$

5.4.1 Derivation

2010: Not included, as the derivation is not part of the curriculum in the course Applied geomatics,

5.4.2 The standard deviation of the weighted mean value

The standard deviation of the weighted mean is obtained by using (5.11):

$$s_x = \hat{\sigma}_x = \frac{\hat{\sigma}_0}{\sqrt{\sum p_i}} = \sqrt{\frac{\sum p_i \hat{\varepsilon}_i^2}{(n-1) \cdot \sum p_i}} = \sqrt{\frac{\sum p_i \hat{v}_i^2}{(n-1) \cdot \sum p_i}} \quad (5.27)$$

Example with numerical values: Weighted mean

Three derived observations of a horizontal angle. The first one is the arithmetical mean value of 5 measurements, the second of 4 and the third of 3 measurements, see chapter 5.3.1. The weights are relative figures and the relations between them are 5 to 4 to 3.

See the tables 5.1 and 5.2.

1. A calculation with 1 single measurement as the unit weight

Observation, l	Weight, p	Estimated correction, $\hat{v}$ in mgon	$p\hat{v}$, control of the sum	$p\hat{v}\hat{v}$
67,12455	5	+0,04	+0,20	0,0080
67,12484	4	-0,25	-1,00	0,2500
67,12432	3	+0,27	+0,81	0,2187
Sum	12		+0,01	0,4767

Table 5.1 The weights are 5, 4, 3

Weighted mean: $\hat{x} = \frac{\sum l_i p_i}{\sum p_i} = 67,12459$ gon

Control of the calculation, see (5.17): $\sum p\hat{v} = 0,01$, which is inside the accuracy of the calculation, shall be 0.

The standard deviation of the unit weight, which is a single measurement (1 of the 12 measurements):

$$s_0 = \sqrt{\frac{\sum p\hat{v}\hat{v}}{n-1}} = \sqrt{\frac{0,4767}{3-1}} = 0,49 \text{ mgon}$$

The standard deviation of the weighted mean: $s_x = \frac{s_0}{\sqrt{\sum p}} = \frac{0,49}{\sqrt{12}} = 0,14 \text{ mgon}$

2. An alternative way of calculation the standard deviations:

The same measurements as in method 1 above, but this time we choose another unit weight:

Choice: The mean value of 5 measurements has the weight equal $p = 1$

The other weights are obtained by using the assumption that the weights are relative figures and thus there must be the same ratios between the three weights in this example, see the table.

Observation, l	Weight, p	Estimated correction, $\hat{v}$ in mgon	$p\hat{v}$, control of the sum	$p\hat{v}\hat{v}$
67,12455	1,0	+0,04	+0,040	0,0016
67,12484	0,8	-0,25	-0,200	0,0500
67,12432	0,6	+0,27	+0,162	0,0438
Sum	2,4		+0,002	0,0954

Table 5.2 The weights are 1,0, 0,8 and 0,6

Weighted mean: $\hat{x} = \frac{\sum l_i p_i}{\sum p_i} = 67,12459$ gon, same as in alternative 1.

Control of the calculation, see (5.17): $\sum p\hat{v} = 0,002$, which is inside the accuracy of the calculation, shall be 0.

The standard deviation of the unit weight, which is the mean value of 5 measurements (5 of the 12 measurements):

$$s_0 = \sqrt{\frac{\sum p\hat{v}\hat{v}}{n-1}} = \sqrt{\frac{0,0954}{3-1}} = 0,22 \text{ mgon}$$

The standard deviation of the weighted mean:

$$s_x = \frac{s_0}{\sqrt{\sum p}} = \frac{0,22}{\sqrt{2,4}} = 0,14 \text{ mgon}, \text{ same as in alternative 1}$$

Then the standard deviation of 1 single measurement is calculated from s_0 :

$s_1 = s_0 \cdot \sqrt{5} = 0,49$ mgon, which is the same value as in alternative 1, where s_0 was the standard deviation of 1 measurement.

5.5 Weights of different observation methods

See the main textbook for weights on different observation methods:

Chapter 9 in “Surveying for Engineers”

See also the lecture notes from Mathematical statistics part 1 and especially part 2

Levelling: $p_{\Delta H} = \frac{p_1 \cdot d}{D} = \frac{k}{D}$

Trigonometrical heighting, long distances (>100 m): $p_{\Delta H} = \frac{k}{D^2}$

Trigonometrical heighting, short distances (<100 m): $p = \text{constant}$

Horizontal directions: $p = \text{constant}$ (if they are measured with the same number of rounds)

Electronic distance meter, standard deviation: $s_D = s_k + s_m \cdot D$. The weights from formula (3.6)

GPS baseline: See lecture notes from Mathematical statistics part 2.